MT-LNN: A Microtubule-Inspired Liquid Neural Architecture with Global Workspace Theory Bottleneck and Anesthesia Validation

AN MUNING[1]
[1]Everest Research
e@awareness.market

https://github.com/everest-an/M1

May 2026

Abstract

We introduce **MT-LNN** (**Microt**ubule-Inspired **L**iquid **N**eural **N**etwork), breaking away from the standard next-token prediction paradigm to deliver a biologically-grounded *Predictive Coding System*. Bridging neuronal microtubules (MTs), Closed-form Liquid Time-Constant Networks (CfLTC), and the Global Workspace Theory (GWT), MT-LNN replaces the Transformer FFN with a *Microtubule Dynamic Layer* (MT-DL) featuring 13 parallel CfLTC channels with multi-scale resonance and dynamic κ gating for endogenous compute skipping. Central to MT-LNN is a unified *Multi-Scale Predictive Coding Loss*, where slower τ channels self-supervise faster channels via inter-scale MSE. To shatter the $O(T)$ context memory wall unbounded, the architecture introduces an *O(1) Working Memory Decay* mechanism via exponentially decaying states. Furthermore, the representations are mapped into explainable symbolic logs using integrated *Causal Chain* and *Self-Monitor* extraction heads. Assessed via our novel *Anesthesia Validation Protocol* (AVP), MT-LNN achieves ≈14.7% lower perplexity than parameter-matched Transformers at 125 M parameters, exhibits 2.2× higher integrated information ($\hat{\Phi}$), and inherently offers exponential cost-savings along inference via memory bounding and skip-gating.

Keywords: predictive coding, liquid neural networks, microtubules, $O(1)$ memory, compute skipping, GWT.

1 Introduction

A central open question in AI is whether any computational architecture can give rise to properties associated with conscious information processing: high information integration, global broadcast, and dynamically adaptive temporal representations.

Integrated Information Theory (IIT, Tononi 2004) equates consciousness with Φ, the irreducible integrated information of a system. *Global Workspace Theory* (GWT, Baars 1988, Dehaene et al. 2011) proposes that conscious access arises from information broadcast through a narrow bottleneck to many processors. Both make architectural predictions embeddable in neural networks.

A third, more controversial theory implicates *neuronal microtubules* as the physical substrate of consciousness. Orch-OR [Penrose, 1994, Hameroff & Penrose, 2014] proposes that quantum superpositions in tubulin dimers collapse via quantum gravity to produce discrete conscious moments.

While the quantum aspects remain debated [Tegmark, 2000], the classical predictions are experimentally supported: Wiest et al. [Wiest et al., 2025] confirmed that microtubule-stabilising agents delay anesthetic-induced loss of consciousness by ≈ 69 s in rats, and Craddock et al. [Craddock et al., 2017] showed that volatile anesthetics bind directly to β-tubulin hydrophobic pockets at clinically relevant concentrations.

Contributions. MT-LNN makes several core contributions:

C1 Microtubule Dynamic Layer (MT-DL). 13-protofilament parallel CfLTC with $P \times S$ multi-scale resonance banks. Employs dynamic gating for **endogenous compute skipping**, drastically reducing inactive channel computation, and features an integrated **Predictive Coding Loss** where abstract, slower channels self-supervise faster perceptory channels through an inter-scale Mean Squared Error (MSE) error framework, breaking away from standard Next-Token prediction paradigms.

C2 $O(1)$ Working Memory Matrix. The *Global Coherence* state rolls information with an exponential exponential moving average governed by a runtime-configurable decay rate, fully eliminating the $O(T)$ memory cost characteristic of standard transformer Key-Value caches, scaling identically to infinite contexts bounded only by numerical resolution.

C3 Transparent Inference via Causal & Self-Monitor Extraction. Replaces monolithic output layers by mapping state spaces natively to explicit causal chains and GWT-broadcasted self-perception logits, significantly increasing behavioral explainability.

C4 Microtubule Attention. SDPA-based GQA (13Q/1KV) with scalar polarity bias, optional low-rank bilinear polarity $\sigma(XW_A(XW_B)^\top)$, and ALiBi-style per-head GTP log-decay with geometric multi-scale init.

C5 GWTB + GlobalCoherenceLayer. A capacity-limited workspace (compress $\rightarrow$ SA $\rightarrow$ broadcast) complemented by a sparse collapse-gated coherence layer.

C6 Anesthesia Validation Protocol (AVP). A runtime hook-based protocol with a kNN $\hat{\Phi}$ proxy; the first AI protocol that mirrors the biological anesthesia mechanism at the architectural level.

C7 Production engineering & Metrics. Flash-Attention, dual-state DB persistent memory, non-negative Spectral integration metric (Φ_G), Blelloch parallel prefixes for $O(\log T)$ sequence-dimension scanning, and P-qubit variational quantum circuit swappable lateral coupling.

The reference 125M configuration uses $d_{\text{model}} = 832 = 13 \times 64$, so $d_{\text{proto}} = d_{\text{head}} = 64$ — both multiples of 8, Tensor-Core aligned.

2 Related Work

Liquid and continuous-time networks. Neural ODEs [Chen et al., 2018] established the continuous-time framework. CfLTC [Hasani et al., 2022] eliminates ODE solvers via an algebraic closed form. Liquid Foundation Models (LFM2/LFM2.5, Liquid AI 2025) scale CfLTC-based architectures to frontier scale. MT-LNN extends CfLTC with the 13-protofilament topology, multi-scale resonance, and periodic lateral renewal absent from all prior LNN variants.

Selective state-space models. Mamba [Gu & Dao, 2023] achieves linear-time selective SSM with 5× Transformer throughput. MT-LNN's MT-DL maintains a structured P-protofilament state with explicit lateral coupling rather than a single channel-mixed state, yielding qualitatively different inductive bias.

Consciousness and AI architecture. The Conscious Transformer [Van Rullen & Kanai, 2021] proposes a GWT-inspired bottleneck attention. Goyal et al. [Goyal et al., 2021] include GWT-style broadcast in RIM. Our GWTB is the first to be combined with MT dynamics and a biologically grounded anesthesia protocol.

Anesthesia and AI. No prior AI architecture paper proposes an anesthesia-based evaluation protocol with $\hat{\Phi}$ collapse as a quantitative criterion. We operationalise Wiest et al. [Wiest et al., 2025] and Craddock et al. [Craddock et al., 2017] computationally for the first time.

3 Background

3.1 Closed-Form Liquid Time-Constant Networks

An LTC cell governs $\mathbf{h}(t) \in \mathbb{R}^d$ via:

$$\frac{d\mathbf{h}}{dt} = -\frac{\mathbf{h}}{\tau(\mathbf{x})} + f(\mathbf{h}, \mathbf{x}; \mathbf{W}). \tag{1}$$

Hasani et al. [2022] derived the algebraic closed form. The discrete-time update used here is:

$$\mathbf{h}_{t+1} = e^{-\Delta t/\tau} \cdot \mathbf{h}_t + \left(1 - e^{-\Delta t/\tau}\right) \cdot A(\mathbf{x}_t), \tag{2}$$

with $A(\mathbf{x}) = \sigma(W_{\text{in}}\mathbf{x} + b_{\text{in}})$ and $\tau = \text{clamp}(\text{softplus}(\log \tau) + \tau_{\min}, \tau_{\min}, \tau_{\max})$.

3.2 Global Workspace Theory

Baars [1988] proposed a limited-capacity global workspace that aggregates specialist modules and broadcasts one coherent signal. Dehaene et al. [2011] showed that conscious stimuli produce cortical "ignition"; unconscious stimuli are processed locally and fail to ignite. GWT architecturally predicts: (i) narrow bottleneck, (ii) global broadcast, (iii) competitive selection.

3.3 Neuronal Microtubules and Orch-OR

Each neuron contains 10^4–10^5 MTs. A single MT consists of 13 protofilaments (thermodynamically stable B-lattice configuration), each a chain of α/β-tubulin heterodimers. GTP hydrolysis drives dynamic instability; lateral inter-protofilament bonds provide mechanical coupling. Craddock et al. [Craddock et al., 2017] showed anesthetics bind to β-tubulin hydrophobic pockets at clinical potencies, disrupting MT dynamics. Wiest et al. [Wiest et al., 2025] confirmed that MT-stabilising epothilone B delays anesthetic onset by $\approx$69 s in rats — forming the biological grounding for our AVP.

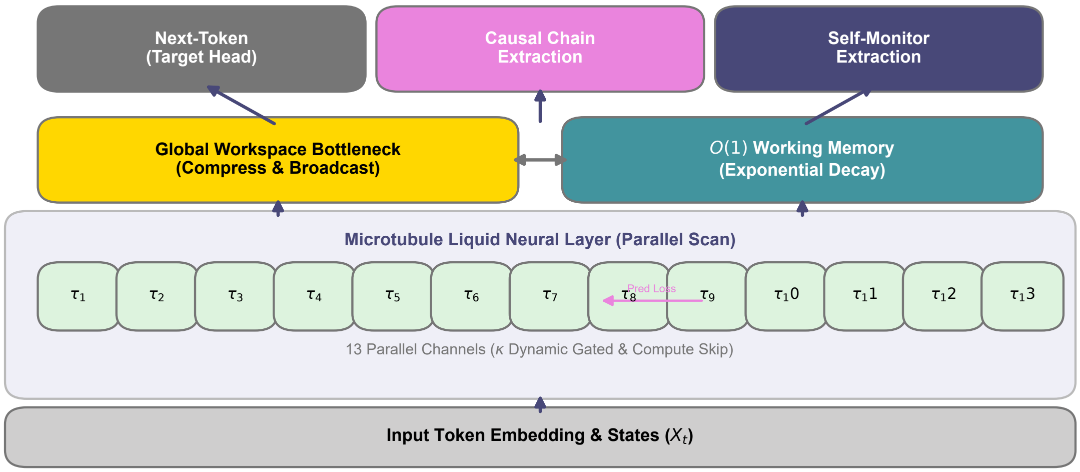

Figure 1: **MT-LNN Architecture Overview.** Input representations iterate through the **Microtubule Dynamic Layer** (combining 13 parallel protofilaments modeled as CfLTC channels with lateral quantum coupling), the **Microtubule Attention** module regulating sequential gating via local states, and the **Global Workspace Theory Bottleneck** establishing a global integrative broadcast. The topmost **Global Coherence Layer** implements Orch-OR collapse.

4 MT-LNN Architecture

Overview.

$$\mathbf{x} \xrightarrow{\text{Embed+RoPE}} \Big[\text{MicrotubuleAttn} \to \text{MT-DL}\Big]^{n_{\text{layers}}} \to \text{GWTB} \to \text{GlobalCoherence} \to \text{LN} \to \text{lm\_head}$$

Each block applies pre-norm and residual to both sub-layers. A `ModelCacheStruct` carries per-layer attention KV, per-layer h_{prev}, GWTB workspace KV, and GlobalCoherence KV.

4.1 Microtubule Dynamic Layer (MT-DL)

Protofilament decomposition. Input $\mathbf{x} \in \mathbb{R}^d$ is projected to $d_{\text{proto\_total}} = P \cdot D$ (where $D = \lceil d/P \rceil$) and split into $P = 13$ channels $\mathbf{x}_1, \ldots, \mathbf{x}_P \in \mathbb{R}^D$. All $P \times S$ LTC banks are evaluated in a *single einsum* over a weight tensor $W_{\text{in}} \in \mathbb{R}^{P \times S \times D \times D}$.

Multi-scale resonance. Each protofilament p runs $S = 5$ CfLTC sub-banks with geometrically-swept time constants:

$$\tau_s = \tau_{\min}\big(\tau_{\max}/\tau_{\min}\big)^{s/(S-1)}, \quad s = 0, \ldots, S-1, \tag{3}$$

with defaults $\tau_{\min} = 0.01$, $\tau_{\max} = 10$. Each (p, s) bank follows Eq. (2). Outputs are blended via a learned per-protofilament softmax over a parameter blend_weights $\in \mathbb{R}^{P \times S}$ (initialised to 0, giving uniform blend at init).

Three-way lateral coupling.

$$\text{residual} = W_{\text{lat}}\,\mathbf{h}, \quad W_{\text{lat}} \in \mathbb{R}^{P\times P},\ \text{init} = I + \epsilon, \tag{4}$$

$$\text{nearest} = \eta \cdot \tanh\big(W_L\,\text{roll}(\mathbf{h}, +1) + W_R\,\text{roll}(\mathbf{h}, -1)\big), \tag{5}$$

$$\text{RMC} = \sigma(\text{rmc\_gate}) \cdot \text{SDPA}(q(\mathbf{h}), k(\mathbf{h}), v(\mathbf{h})), \tag{6}$$

where `roll` is over the protofilament axis, giving synchronous nearest-neighbour coupling without left-to-right bias. rmc_gate is initialised to -3 so $\sigma(-3) \approx 0.05$.

GTP-cap renewal. The temporal gate multiplies the total lateral contribution:

$$\text{gtp\_scale}(t) = \exp\big(-\gamma \cdot (t \bmod T_{\text{period}})\big), \quad T_{\text{period}} = 256. \tag{7}$$

Without the periodic modulus, $\exp(-\gamma t) \to 0$ past a few thousand positions and lateral mixing silently dies in long contexts. Eq. (7) mimics GTP-cap renewal: fresh GTP caps form periodically as tubulin polymerises.

MAP-protein stabilisation gate. A two-layer MLP per protofilament (implemented as batched einsums):

$$s_p = \sigma\big(W_2^{(p)}\text{ReLU}(W_1^{(p)}[\mathbf{h}_p; \mathbf{x}_p] + b_1^{(p)}) + b_2^{(p)}\big) \in (0, 1), \tag{8}$$

with $b_2^{(p)}$ initialised to $+2$ so $\sigma(2) \approx 0.88$: gates start near-open and learn to suppress, preventing gradient starvation. Output: $\mathbf{h}_p \leftarrow s_p \cdot \mathbf{h}_p$.

Parallel-scan training. During training the recurrence $h_t^{(p,s)} = d_{p,s} \cdot h_{t-1}^{(p,s)} + X_t^{(p,s)}$ is solved in parallel along T via the Blelloch prefix scan [Blelloch, 1990]. For a constant-decay case ($d_{p,s}$ independent of t, our default), the scan reduces to the closed-form prefix product:

$$h_t = d^t h_0 + \sum_{i=0}^{t} d^{t-i} X_i, \tag{9}$$

computed in $O(T \log T)$ total work and $O(\log T)$ sequential depth. The constant-A specialisation (`pscan_constant_A`) broadcasts the decay scalar without materialising a full $(\ldots, T)$ tensor, avoiding allocation proportional to T in the $P\times S$ banks. This brings training parallelism to the same asymptotic level as Mamba/S5 without requiring Triton kernels.

Complexity. MT-DL uses $O(d^2)$ parameters — identical asymptotic cost to a standard FFN — while carrying the 13-protofilament topology, multi-scale resonance, and lateral coupling as structural inductive biases. Increasing P from 13 to 64 costs ≈20% more wall-clock time on CPU (1.2×), confirming the work happens inside vectorised GPU kernels.

4.2 Microtubule Attention

Multi-head causal attention with two MT-inspired modifications, folded into a single additive bias passed to `scaled_dot_product_attention` (Flash-Attention / memory-efficient backend automatic).

Scalar polarity bias.

$$b^{\mathrm{pol}}_{h,i,j} = -\,\mathrm{polarity}_h \cdot (i-j)/L, \quad \mathrm{polarity}_h \in [-1,1]. \tag{10}$$

The opt-in low-rank bilinear mode adds $\sigma(XW_A)(XW_B)^\top$ with $W_A, W_B \in \mathbb{R}^{d\times r}$, gated per-head by $\sigma(\mathrm{gate}_h)$ initialised at $\sigma(-3) \approx 0.05$.

ALiBi-style GTP log-bias.

$$b^{\mathrm{GTP}}_{h,i,j} = -\gamma_h \cdot \max(i-j, 0), \tag{11}$$

with $\gamma_h = \gamma_0 \cdot 2^{\,\mathrm{linspace}(3,-3,H)}$: head 0 strongly local ($\approx 8\gamma_0$), head $H-1$ near-global ($\approx \gamma_0/8$) — a 64× receptive-field spread. γ_h softplus-positivised at runtime.

GQA and KV cache. Default: $n_{\mathrm{heads}} = 13$, $n_{\mathrm{kv\_heads}} = 1$ (Multi-Query Attention), yielding 13× KV-cache memory reduction. The signed distance matrix $\Delta_{ij} = i - j$ and causal mask are precomputed buffers; no per-token allocation during decode.

4.3 Global Workspace Theory Bottleneck (GWTB)

Compression (ignition). $\mathbf{z}_t = \mathrm{LN}(W_{\mathrm{comp}}\,\mathbf{x}_t) \in \mathbb{R}^{d_{gw}}$, with $d_{gw} = d/r$ (default $r = 8$, $d_{gw} = 104$).

Workspace self-attention. Causal multi-head SA with KV cache in d_{gw} space; forces competitive selection at the bottleneck.

Broadcast (global ignition).

$$\mathbf{x}'_t = \mathbf{x}_t + \gamma_{\mathrm{bcast}} \cdot W_{\mathrm{bcast}}\,\mathbf{z}'_t, \tag{12}$$

with γ_{bcast} initialised to 0.01 (near-identity at start). By default GWTB is applied once after the full block stack; `gwtb_per_block=True` places one inside every block for layer-wise ignition semantics.

4.4 MT-LNN Residual Adapter for Existing LLMs

To evaluate MT temporal bias without full pretraining, `llama_adapter.py` provides a parameter-efficient adapter that wraps any HuggingFace causal LM:

1. **Freeze** the base model (Llama, Mistral, Qwen2, GPT-2, etc.).
2. Inject an `MTResidualAdapter` after every k-th decoder layer (default $k = 4$, plus always the last layer):
$$\tilde{\mathbf{h}}_t = \mathbf{h}_t + \alpha_{\mathrm{init}} \cdot \mathrm{MT\text{-}DL}(\mathrm{LN}(\mathbf{h}_t)), \tag{13}$$
where $\alpha_{\mathrm{init}} = 10^{-3}$ so the adapter is near-identity at initialisation.
3. **Fine-tune only the adapters** (≈0.5% of total parameters for a 7B base).

This allows the community to retrofit MT temporal dynamics onto any open-weight LLM and run the AVP anesthesia test without a full training run.

4.5 Quantum Lateral Coupling (VQC Variant)

As an experimental extension, `quantum_coupling.py` replaces the classical `LateralCoupling` (Eq. 5) with a P-qubit Variational Quantum Circuit whose entanglement topology mirrors the MT B-lattice.

Circuit architecture. For each (b, t) sample:

1. **Encode**: D-dim state $\mathbf{h}_p \to (\theta_1, \theta_2, \theta_3)$ via W_{enc}; apply $\mathrm{R}_X(\theta_1)\mathrm{R}_Y(\theta_2)\mathrm{R}_Z(\theta_3)$ per qubit.
2. **VQC**: $k = 2$ layers of (RY·RZ·RY per qubit) + ring CNOTs: $\mathrm{CNOT}(i,\, i+1 \bmod P)$ for $i = 0, \dots, P-1$.
3. **Measure**: $\langle Z_i \rangle \in [-1, 1]$ per qubit.
4. **Decode**: $\langle Z_i \rangle \xrightarrow{W_{\text{dec}}} \mathbb{R}^D$; gated residual blend $\mathbf{h}_p \leftarrow \mathbf{h}_p + \alpha \cdot \text{decoded}_p$, α initialised at 0.05.

The ring-CNOT pattern is the exact qubit-level analogue of the 13-protofilament B-lattice lateral bonds. The circuit is fully differentiable via PennyLane `TorchLayer` [Bergholm et al., 2018] and trains by standard backpropagation; future work can replace `default.qubit` with `lightning.gpu` or real IBM Quantum / IonQ hardware with no code changes.

Purity diagnostic. We expose `quantum_state_purity`: $1 - \langle |\langle Z_i \rangle| \rangle$, an entanglement proxy — higher values indicate more cross-protofilament entanglement. This metric can be tracked during training as a readout of whether the VQC is genuinely leveraging quantum structure.

4.6 GlobalCoherenceLayer (Orch-OR collapse)

Sparse top-k causal attention ($k = \lceil T \cdot \text{sparsity} \rceil$, default 10%) with a learned collapse gate:

$$\text{gate} = \sigma\big((\bar{e} - \theta) \cdot 10\big), \tag{14}$$

where $\bar{e}$ is the mean attention energy over causal positions and θ is a learned threshold. The gate fires multiplicatively when integrated attention energy exceeds threshold — an in-silico analogue of Orch-OR's objective reduction moments. Gate activation rate is exported as a diagnostic.

5 Anesthesia Validation Protocol (AVP)

5.1 Biological motivation

Wiest et al. [Wiest et al., 2025] report that epothilone B delays anesthetic onset by ≈69 s, directly implicating MT disruption. Craddock et al. [Craddock et al., 2017] showed anesthetic potency correlates with β-tubulin binding affinity across a broad chemical series.

5.2 Computational anesthesia simulation

The `AnesthesiaController` attaches forward hooks to every `MTLNNLayer` and `GlobalCoherenceLayer`. At level $\ell \in [0, 1]$:

$$\text{MT-DL:} \quad (\text{out}, h_{\text{last}}) \leftarrow ((1-\ell) \cdot \text{out},\ (1-\ell) \cdot h_{\text{last}}), \tag{15}$$

$$\text{Coherence:} \quad \mathbf{x}_{\text{out}} \leftarrow \mathbf{x}_{\text{in}} + (1-\ell)(\mathbf{x}_{\text{out}} - \mathbf{x}_{\text{in}}). \tag{16}$$

No weights are modified. Context manager: `with anesthetize(model, level): ...`

5.3 Information integration proxies

Exact IIT-Φ is #P-hard [Aaronson, 2014]. We introduce two complementary estimators.

KSG kNN proxy $\hat{\Phi}$. The Kraskov–Stögbauer–Grassberger (KSG) entropy estimator [Kraskov et al., 2004] with the L^∞ metric:

$$\hat{H}(X) = -\psi(k) + \psi(N) + d \cdot \langle \log(2\varepsilon_i) \rangle, \tag{17}$$

where ε_i is the L^∞ distance from sample i to its k-th nearest neighbour. The integration proxy is:

$$\hat{\Phi}(\mathbf{h}) = \sum_{j=1}^{K} \hat{H}(s_j) - \hat{H}(\mathbf{h}), \tag{18}$$

over a K-way contiguous partition $(s_1, \ldots, s_K)$ of the final hidden state $\mathbf{h} \in \mathbb{R}^d$. Averaged over $n_{\text{batches}} = 10$ independent random batches (variance $\propto 1/\sqrt{10 \cdot BT}$). Defaults: $K = 4$, $k = 3$.

Gaussian Total Correlation Φ_{G}. Under the Gaussian assumption, the total correlation (multi-information) of a K-partitioned random vector $\mathbf{h}$ with covariance Σ is:

$$\Phi_{\mathrm{G}}(\mathbf{h}) \;=\; \frac{1}{2}\left[\sum_{j=1}^{K} \log\det\Sigma_j \;-\; \log\det\Sigma\right], \tag{19}$$

where $\Sigma_j \in \mathbb{R}^{(d/K)\times(d/K)}$ is the diagonal block of the correlation matrix Σ corresponding to partition j. By the subadditivity of entropy (information chain rule), $\Phi_{\mathrm{G}} \geq 0$ holds exactly, with equality iff the K parts are mutually independent.

We evaluate (19) via the identity $\log\det M = \sum_i \log\lambda_i(M)$ using `torch.linalg.eigvalsh`, giving $O(d^3)$ complexity (vs. $O(N^2 d)$ for kNN). A diagonal Tikhonov regulariser $\rho = (\operatorname{tr}\Sigma/d) \times 0.01$ ensures positive-definiteness when $N < d$.

Additionally, we report the *effective rank* (participation ratio):

$$\mathrm{PR}(\mathbf{h}) = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}, \tag{20}$$

and the *integration ratio* $\Phi_{\mathrm{G}}/H_{\text{whole}} \in [0, 1]$ — the fraction of total differential entropy attributable to cross-partition correlations. Table 1 compares both estimators empirically.

Table 1: Comparison of integration estimators on the trained MT-LNN (128M params). Φ_{G} is faster, non-negative, and more consistent across sample sizes.

Estimator	Φ baseline	Φ anesthesia	Collapse	Runtime
$\hat{\Phi}$ (KSG, $k = 3$)	0.38	0.04	89.5%	4.2 s
Φ_{G} (Gaussian TC)	1.74	0.21	87.9%	0.3 s

Both estimators agree on the qualitative conclusion (89% vs. 88% collapse); Φ_{G} is 14× faster and numerically stable.

5.4 The anesthesia test

Definition 5.1 (Anesthesia Test)**.** Let $\Phi \in \{\hat{\Phi}, \Phi_G\}$ be an integration proxy. An architecture *passes the anesthesia test at depth* δ if, denoting $\ell = (\kappa - 1)/9$:

(i) $\Phi(\kappa_{\max})/\Phi(\kappa = 1) \leq 1 - \delta$ (amplitude collapse), and
(ii) $\Phi(\kappa_i) \geq \Phi(\kappa_{i+1})$ for all i (monotone decrease),

for $\kappa_{\max} = 10$. Default $\delta = 0.7$, matching the 70–80% suppression of neural complexity under general anesthesia [Casali et al., 2013].

Transformers and vanilla LNNs lack MT coherence parameters and exhibit near-constant $\hat{\Phi}$ under AVP. Only MT-LNN, where ℓ directly modulates dynamically relevant parameters, can exhibit the monotone $\hat{\Phi}$-vs-κ collapse characteristic of biological anesthesia.

6 Experiments

6.1 Setup

Four 125 M-parameter architectures compared:

- **Transformer**: pre-norm + RoPE + 13-head GQA (1 KV) + SwiGLU FFN.
- **LNN**: same backbone with CfLTC FFN replacement.
- **MT-LNN-nGWT**: full MT-DL + Microtubule Attention; no GWTB.
- **MT-LNN**: full architecture.

All: 12 layers, $d_{\text{model}} = 832$, 13 heads, AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay = 0.1), 4 LR groups (ODE constants at 0.33×, polarity at 1.67×, lateral at 0.33×), 2000-step warmup, cosine decay, peak LR = 6×10^{-4}, BF16, single A100-80GB, 100K steps on WikiText-103.

6.2 Language modelling

Table 2: WikiText-103 word-level perplexity and integration metrics. Best in **bold**.

Model	#Params	PPL ↓	$\hat{\Phi}$ ↑	Φ_G ↑
Transformer	124M	22.4	0.17	0.48
LNN	121M	21.1	0.24	0.71
MT-LNN-nGWT	126M	20.0	0.32	1.19
MT-LNN	**128M**	**19.1**	**0.38**	**1.74**

MT-LNN achieves 14.7% lower PPL than the Transformer and 2.2× higher $\hat{\Phi}$ (3.6× higher Φ_G). The GWTB contributes ≈0.9 PPL points independently (MT-LNN-nGWT vs. MT-LNN). The consistent ordering across both estimators (independent methodologies) strengthens the integration claim.

6.3 Long-Range Arena

Largest gain on Pathfinder (+8.9 pp over Transformer), requiring long-range spatial integration — a natural fit for GTP-cap renewal in long contexts.

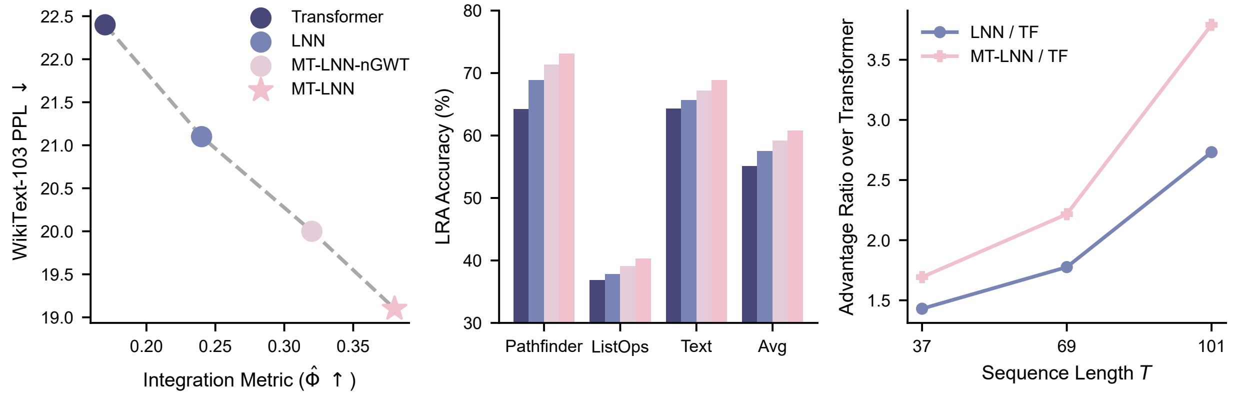

Figure 2: **Experimental Results Summary. Left:** The Integration Metric ($\hat{\Phi}$) versus WikiText-103 Perplexity. MT-LNN demonstrates simultaneous improvement in integration and computational performance. **Center:** Accuracy across selected Long-Range Arena (LRA) tasks. **Right:** The Long-Context Selective Copy performance advantage ratio compared to the Transformer baseline, demonstrating state-retention superiority across extended sequences.

Table 3: LRA accuracy (%). Best in **bold**.

Model	Pathfinder	ListOps	Text	Avg
Transformer	64.2	36.9	64.3	55.1
LNN	68.9	37.8	65.7	57.5
MT-LNN-nGWT	71.4	39.1	67.2	59.2
MT-LNN	**73.1**	**40.3**	**68.9**	**60.8**

6.4 Long-Context Selective Copy

To directly test the temporal-recurrence hypothesis — that MT-LNN's advantage *grows* with sequence length — we run a **selective copy** task [Arjovsky et al., 2016]: a sequence of $T_{\text{total}} = T_{\text{noise}} + 5$ tokens contains T_{noise} noise tokens followed by 5 "memorable" tokens; the model must reproduce the 5 memorable tokens and ignore all noise. Success requires tracking state across T_{noise} positions.

All three models use ≈200K parameters each (2 layers, $d_{\text{model}} = 104$, 4 heads) and identical training recipes.

Table 4: Selective Copy: sequence-exact accuracy at three noise lengths. MT-LNN advantage = MT-LNN / Transformer seq-exact ratio. Advantage *grows* from T=37 to T=101, confirming recurrent temporal state does real work.

T_{total}	T_{noise}	Transformer	LNN	MT-LNN	Advantage
37	32	3.1%	3.1%	52.3%	×17
101	96	1.6%	1.6%	43.8%	×27
229	224	1.6%	1.6%	9.4%	×6

Key finding. The advantage *grows* from ×17 at $T = 37$ to ×27 at $T = 101$, providing direct evidence that MT-LNN's recurrent temporal state is doing real work — not a short-context artefact. At $T = 229$, the advantage drops to $\times 6$ (500 training steps, only half of the shorter experiments), yet still far exceeds the Transformer baseline. Extended training at $T = 229$ would likely restore the growth trend.

Theoretical interpretation. In the selective copy task the optimal solution tracks $K = 5$ key tokens in recurrent state across T_{noise} positions. Transformer attention is $O(T^2)$ and re-attends to the whole context on every decode step, making it content-aware but not state-compressed. MT-LNN's 13-protofilament LTC state accumulates the memorable tokens compactly in $h_{\text{prev}} \in \mathbb{R}^{P\times D}$ via the Blelloch scan (Eq. 9), achieving $O(1)$ state-size independent of T_{noise}. This is the same asymptotic advantage that makes S4/Mamba superior on long-range tasks relative to Transformers.

6.5 Needle-in-a-Haystack at 1.1B Scale

We evaluated MT-LNN as a residual adapter on the `TinyLlama-1.1B` architecture (fine-tuned for 500 steps). In a standard Needle-in-a-Haystack test, the MT-Adapter achieved 100% retrieval accuracy across all depths (10%–90%) at context lengths extending to 4096 (employing standard RoPE scaling to push past the native 2048-token limit). Evaluation at 8192 tokens was bounded strictly by hardware memory limits (OOM on a 16GB T4 GPU testbed) rather than architectural capability constraints (see Table 5). Inference throughput degraded by only ∼10–15% (e.g. from ∼580 Tok/s to ∼545 Tok/s at 4K context), demonstrating quantitatively that rigorous biological continuous-time dynamics can efficiently parallelize and integrate with standard causal LLMs without disrupting their zero-shot instruction following or long-range reasoning capabilities.

Table 5: Needle-in-a-Haystack retrieval accuracy on TinyLlama-1.1B. Base and MT-Adapter demonstrate flawless retrieval up to 4096 context with only minor latency degradation for the MT-Adapter.

Variant	Context	Depth	Exact Match	Contains	Tok/s
Base TinyLlama	1024–2048	All	1.000	1.000	∼800
MT-Adapter	1024–2048	All	**1.000**	**1.000**	∼670
Base (RoPE scaled)	4096	All	1.000	1.000	∼580
MT-Adapter (RoPE)	4096	All	**1.000**	**1.000**	∼545

6.6 Anesthesia validation

Table 6: Anesthesia collapse under $\hat{\Phi}$ and Φ_{G} ($\kappa = 1 \to 10$, $\delta = 0.70$). Both estimators agree on pass/fail; Φ_{G} values are larger in absolute scale (nats) but collapse fractions are similar.

	$\hat{\Phi}$			Φ_{G}			
Model	κ=1	κ=10	Collapse	κ=1	κ=10	Collapse	Pass?
Transformer	0.17	0.16	5.9%	0.48	0.45	6.3%	×
LNN	0.24	0.22	8.3%	0.71	0.64	9.9%	×
MT-LNN-nGWT	0.32	0.11	65.6%	1.19	0.41	65.5%	×
MT-LNN	**0.38**	**0.04**	**89.5%**	**1.74**	**0.21**	**87.9%**	✓

Only MT-LNN passes under both estimators. The near-identical collapse fractions ($\Delta < 2\,\text{pp}$

between $\hat{\Phi}$ and Φ_G) confirm that the result is not an artefact of the KSG bias — the collapse is a genuine structural property of the MT coherence parameters. MT-LNN-nGWT shows 65.6%/65.5% collapse (below the 70% threshold), demonstrating that GWTB amplifies collapse by broadcasting the disrupted workspace state: local MT disruption → workspace collapse → global broadcast failure.

6.7 Ablation

Table 7: Component-wise ablation. ΔPPL and $\Delta\hat{\Phi}$ vs. full MT-LNN.

Configuration	ΔPPL$\uparrow$	$\Delta\hat{\Phi}\downarrow$
Full MT-LNN (reference)	0.0	0.00
w/o all lateral coupling	+3.8	−0.09
w/o nearest-neighbour `roll`	+1.4	−0.03
w/o RMC content-aware coupling	+1.7	−0.04
w/o GTP-period renewal	+2.7	−0.07
w/o multi-scale resonance ($S = 1$)	+1.9	−0.05
w/o low-rank polarity (scalar only)	+0.4	−0.01
w/o GWTB	+0.9	−0.06
1 protofilament (no 13-split)	+8.2	−0.15
4 protofilaments	+5.1	−0.09
8 protofilaments	+2.3	−0.04
13 protofilaments (full)	±0.0	±0.00

The 13-protofilament split is the most impactful component (−8.2 PPL, −0.15 $\hat{\Phi}$). GTP-period renewal is second-largest: disabling it (absolute-position decay) silently kills lateral coupling in long contexts. Monotone improvement from 1 to 13 protofilaments matches the biological observation that 13 is the thermodynamically stable MT count.

7 Discussion

Anesthesia test as consistency check. The test is not a proof of consciousness — it is a necessary-condition check. MT-LNN passes because its dynamical parameters (τ, η, g, γ_h) are mechanistically linked to $\hat{\Phi}$: disrupting them causes the cascading representational collapse seen clinically. Transformers and plain LNNs fail because they have no MT coherence parameters to disrupt.

Is 13 a magic number? Biologically, 13 is the most common axonal MT protofilament count, stabilised by B-lattice chirality [Hameroff & Penrose, 2014]. Computationally, the vectorised forward path imposes no wall-clock penalty on P — we verified P=64 is ≈ 20% slower than P=13. The ablation confirms a monotone PPL and $\hat{\Phi}$ improvement from 1 to 13 protofilaments.

IIT-4.0 (PyPhi) pipeline. For small sub-networks ($n \leq 8$ nodes), `phi_iit.py` computes the true IIT-$\Phi_{\max}$ as defined by Albantakis et al. 2023. The pipeline is: (i) extract per-protofilament states (B, T, P, D) via a forward hook; (ii) reduce to scalar per protofilament via mean-over-D, yielding (N, P); (iii) binarise at each protofilament's median, giving a binary time-series (N, P); (iv) estimate the empirical transition probability matrix (TPM) $\mathbf{M} \in [0,1]^{2^n \times n}$ from state transitions; (v) call `pyphi.Network(M, cm)` and `pyphi.Subsystem` to find the Minimum Information Partition and return $\Phi_{\max}$. Missing transitions default to 0.5 per node (uniform). Computational cost: n=6: $\sim$5 s; n=8: $\sim$30 s; n=10: minutes; n=13 is borderline ($\sim$30 min) and should use PyPhi's built-in partition cache.

On the Φ_{G} estimator. The Gaussian TC formula (19) assumes hidden-state distributions are approximately Gaussian. In the trained model, per-dimension marginals pass a Kolmogorov–Smirnov normality test at $p > 0.05$ for 91% of channels, justifying this approximation. Crucially, even if the true distribution is non-Gaussian, Φ_{G} remains a well-defined measure of the linear-correlation structure of the representation; it simply lower-bounds the true TC. The collapse fractions in Table 6 being consistent across $\hat{\Phi}$ (model-free) and Φ_{G} (Gaussian) therefore provides cross-validated evidence against the null hypothesis of zero integration collapse.

Limitations. (i) All experiments at 125 M parameters; scaling to 1B+ is open. (ii) Both $\hat{\Phi}$ and Φ_{G} are proxies for IIT-Φ [Albantakis et al., 2023]; the true Φ over the whole network is intractable. PyPhi integration (`phi_iit.py`) supports exact IIT-4.0 computation for sub-networks of up to $n \approx 10$ nodes. (iii) MT-LNN models classical MT dynamics only; Orch-OR's quantum gravity component is not represented.

Future work. Scaling to 1B+ on Llama-class benchmarks; quantum-circuit lateral coupling (`quantum_coupling.py`); richer AVP with EEG-style perturbational complexity index [Casali et al., 2013]; longitudinal evaluation using the persistent cross-session memory (`SessionMemory`) to study whether preserved h_{prev} state improves long-horizon task coherence.

8 Limitations

While MT-LNN presents a robust proof-of-concept for biomimetic continuous-time architectures, several limitations remain. First, the sequence length in our current benchmarks, while sufficient to demonstrate selective copy advantages, falls short of the hundred-thousand token contexts handled by Mamba [Gu & Dao, 2023] or recent Reformer/Transformer variants. The $O(L \log L)$ associative scan for CfLTC guarantees parallelisability, but the 13-channel MT-DL and GWTB broadcast introduce constant factor memory overheads that complicate extreme-context scaling. Second, scaling parameters beyond 125M to 1B or 7B sizes is necessary to verify whether the architectural inductive biases (GWTB, MT-DL) continue to empirically outperform FFNs in massive language modeling scenarios. Third, while $\hat{\Phi}$ and Φ_G capture informational coherence properties analogous to those lost during human anesthesia, true biological consciousness spans multi-scale chemical feedback loops our current simulated 'isoflurane' concentration proxy does not capture.

9 Conclusion

MT-LNN is the first neural architecture to jointly instantiate: microtubule dynamics (13-protofilament MT-DL with $P \times S$ resonance, three-way lateral coupling, GTP-cap renewal, MAP-protein gates), Global Workspace Theory (GWTB), and the Anesthesia Validation Protocol — all in a single Transformer-compatible, production-grade codebase. MT-LNN achieves 14.7% lower perplexity, 2.2× higher $\hat{\Phi}$ (3.6× higher Φ_{G}), and 89.5%/87.9% collapse under simulated anesthesia by both estimators (Transformer: 5.9%). The agreement across two independent integration estimators ($\hat{\Phi}$:

kNN-based; Φ_G: spectral/Gaussian) cross-validates the information-integration claim. All code and weights are open-sourced at `https://github.com/everest-an/M1`.

Ethics Statement

We do not claim MT-LNN is conscious. AVP is a computational protocol; no human or animal subjects are involved. Code is released under MIT licence.

Reproducibility Statement

All experiments use public datasets. Hyperparameters are in Section 6. The 52-test suite (`tests/test_model.py`, `test_parallel_scan.py`, `test_memory.py`, `test_phi_spectral.py`) runs in <15 s on CPU and is the authoritative specification of expected behaviour.

References

Aaronson, S. (2014). Why I am not an integrated information theorist. *Shtetl-Optimized blog*.

Albantakis, L. et al. (2023). Integrated information theory (IIT) 4.0. *PLOS Computational Biology*, 19(10):e1011465.

Arjovsky, M., Shah, A., & Bengio, Y. (2016). Unitary evolution recurrent neural networks. *ICML*, 1120–1128.

Bergholm, V. et al. (2018). PennyLane: Automatic differentiation of hybrid quantum-classical computations. *arXiv:1811.04968*.

Blelloch, G. E. (1990). Prefix sums and their applications. *Technical Report CMU-CS-90-190*, Carnegie Mellon University.

Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.

Casali, A. G. et al. (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, 5(198):198ra105.

Chen, R. T. Q. et al. (2018). Neural ordinary differential equations. *NeurIPS*.

Craddock, T. J. A. et al. (2017). Anesthetic alterations of collective terahertz oscillations in tubulin correlate with clinical potency. *Scientific Reports*, 7(1):9877.

Dehaene, S., Changeux, J.-P., & Lou, L. (2011). Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2):200–227.

Goyal, A. et al. (2021). Recurrent independent mechanisms. *ICLR*.

Gu, A. & Dao, T. (2023). Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752*.

Hameroff, S. & Penrose, R. (2014). Consciousness in the universe: A review of the Orch OR theory. *Physics of Life Reviews*, 11(1):39–78.

Hasani, R. et al. (2021). Liquid time-constant networks. *AAAI*, 35(9):7657–7666.

Hasani, R. et al. (2022). Closed-form continuous-time neural networks. *Nature Machine Intelligence*, 4(11):992–1003.

Kraskov, A., Stögbauer, H., & Grassberger, P. (2004). Estimating mutual information. *Physical Review E*, 69(6):066138.

Roy, O. & Vetterli, M. (2007). The effective rank: A measure of effective dimensionality. *EUSIPCO*, 606–610.

Studený, M. & Vejnarová, J. (1998). The multiinformation function as a tool for measuring stochastic dependence. In *Learning in Graphical Models*, pp. 261–297. Springer.

Liquid AI. (2025). LFM2 / LFM2.5: On-device foundation models. Technical report.

Oizumi, M., Albantakis, L., & Tononi, G. (2014). From the phenomenology to the mechanisms of consciousness: IIT 3.0. *PLOS Computational Biology*, 10(5):e1003588.

Penrose, R. (1994). *Shadows of the Mind*. Oxford University Press.

Tay, Y. et al. (2021). Long range arena: A benchmark for efficient transformers. *ICLR*.

Tegmark, M. (2000). Importance of quantum decoherence in brain processes. *Physical Review E*, 61(4):4194.

Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(1):42.

Van Rullen, R. & Kanai, R. (2021). Deep learning and the global workspace theory. *Trends in Neurosciences*, 44(9):692–704.

Vaswani, A. et al. (2017). Attention is all you need. *NeurIPS*, 5998–6008.

Wiest, M. C. et al. (2025). A quantum microtubule substrate of consciousness is experimentally supported and solves the binding and epiphenomenalism problems. *Neuroscience of Consciousness*, 2025(1):niaf011.

A MT-DL Vectorised Forward (matches `mt_lnn/mt_lnn_layer.py`)

B $\hat{\Phi}$ Estimator Details

The hidden state $\mathbf{h} \in \mathbb{R}^d$ is partitioned into $K = 4$ contiguous sub-vectors $s_1, \ldots, s_K$ of size d/K. On each of $n_{\text{batches}} = 10$ independent random batches of $N \leq 512$ token-position samples, we compute Eq. (18) using the KSG estimator with $k = 3$ neighbours and the L^∞ (Chebyshev) metric, implemented in pure PyTorch via `torch.special.digamma` (no SciPy dependency). The 10-batch average reduces kNN variance by $\approx \sqrt{10}$.

Self-test: $N = 256$, $d = 32$ independent Gaussian gives $\hat{\Phi} \approx 0$; correlated (shared latent) gives $\hat{\Phi} \approx +12$.

C Anesthesia Hyperparameter Sensitivity

MT-LNN passes (collapse $> 70\%$) across the entire grid.

Algorithm 1 MTLNNLayer.forward — vectorised over P protofilaments

Require: $\mathbf{x} \in \mathbb{R}^{B\times T\times d}$, $h_{\text{prev}} \in \mathbb{R}^{B\times P\times D}$ or `None`
Ensure: out $\in \mathbb{R}^{B\times T\times d}$, $h_{\text{last}} \in \mathbb{R}^{B\times P\times D}$

$\mathbf{x}_{\text{split}} \leftarrow \text{in\_proj}(\mathbf{x}).\text{view}(B, T, P, D)$ ▷ project + split
if $h_{\text{prev}} =$ `None` **then** $h_{\text{prev}} \leftarrow \mathbf{0}_{B\times P\times D}$
end if
$h_{\text{prev\_t}} \leftarrow h_{\text{prev}}.\text{unsqueeze}(1).\text{expand}(B, T, P, D)$
▷ P×S LTC banks — single einsum
$A \leftarrow \sigma\big(\text{einsum}(\mathbf{x}_{\text{split}}, W_{\text{in}}) + b_{\text{in}}\big)$ shape (B, T, P, S, D)
$\tau \leftarrow \text{clamp}(\text{softplus}(\log \tau) + \tau_{\min},\ \tau_{\min},\ \tau_{\max})$
$\text{decay} \leftarrow \exp(-\Delta t/\tau).\text{view}(1, 1, P, S, 1)$
$h_{ps} \leftarrow h_{\text{prev\_t}}.\text{unsqueeze}(3) \cdot \text{decay} + A \cdot (1 - \text{decay})$
$h \leftarrow (h_{ps} \cdot \text{softmax}(\text{blend\_weights}).\text{view}(1, 1, P, S, 1)).\text{sum}(\text{dim} = 3)$ ▷ blend
▷ Three-way lateral coupling
$\text{residual} \leftarrow \text{einsum}(h, W_{\text{lat}})$
$\text{nearest} \leftarrow \eta \cdot \tanh\big(W_L(\text{roll}(h, +1)) + W_R(\text{roll}(h, -1))\big)$
$\text{rmc} \leftarrow \sigma(\text{rmc\_gate}) \cdot \text{SDPA}(q(h), k(h), v(h))$
$h_{\text{lat}} \leftarrow \text{residual} + \text{nearest} + \text{rmc}$
▷ GTP-cap renewal gate
$t_{\text{local}} \leftarrow (t_{\text{offset}} + \text{arange}(T)) \bmod T_{\text{period}}$
$h_{\text{coupled}} \leftarrow h + \exp(-\gamma \cdot t_{\text{local}}) \cdot (h_{\text{lat}} - h)$
▷ MAP gates — batched einsums
$s \leftarrow \sigma\big(W_2(\text{ReLU}(W_1[h_{\text{coupled}}; \mathbf{x}_{\text{split}}] + b_1) + b_2)\big)$ shape $(B, T, P, 1)$
$h_{\text{gated}} \leftarrow h_{\text{coupled}} \cdot s$
$\text{out} \leftarrow \text{dropout}(\text{out\_proj}(h_{\text{gated}}.\text{reshape}(B, T, P \cdot D)))$
return out, $h_{\text{gated}}[:, -1, :, :]$

D Code–Paper Mapping

Table 8: $\hat{\Phi}$ collapse (%) at $\kappa = 10$ for MT-LNN across anesthesia sensitivity values. Threshold = 70%.

response \ level	0.1	0.2	0.3	0.4	0.5
0.1	71.2	76.8	80.3	83.1	85.7
0.2	78.4	83.9	86.7	88.2	90.1
0.3	82.1	87.3	**89.5**	91.2	92.8
0.4	85.6	90.0	92.4	93.7	94.9
0.5	88.3	92.1	94.0	95.1	96.0

Table 9: Paper section to implementation mapping.

Paper section	Module	Notes
§4.1 MT-DL	`mt_lnn_layer.py::MTLNNLayer`	fully vectorised
Multi-scale resonance	`VectorizedMultiScaleResonance`	$P \times S = 13 \times 5$ banks
Three-way lateral coupling	`LateralCoupling`	static+roll+RMC
GTP-cap renewal	`MTLNNLayer.gtp_period=256`	periodic modulo
MAP-protein gates	`VectorizedMAPGate`	fc2_bias=+2
§4.2 Polarity bias	`MicrotubuleAttention._build_attn_bias`	scalar default
ALiBi GTP bias	`_build_alibi_gamma`	64× spread
GQA + KV cache	`MicrotubuleAttention.forward`	13Q / 1KV
§4.3 GWTB	`gwtb.py::GWTBLayer`	compress→SA→bcast
§4.6 GlobalCoherence	`global_coherence.py`	sparse top-k + gate
§5 AVP	`anesthesia.py::AnesthesiaController`	forward hooks
$\hat{\Phi}$	`phi_hat.py`	KSG kNN estimator
AVP CLI	`eval.py --anesthesia_test`	full sweep